04c — Polar vs rectangular: why two forms?

Follows 04b · a maths doubt that AC circuits force on you · textbook: Sadiku ch. 9 §9.3 + App. B (complex numbers) · every number checked by script (`verify_polar.py`)

Map

04b: phasor = size∠angle → **1 Same arrow, two addresses** → 2 Adding: rectangular wins →
3 Multiplying: polar wins → 4 Why angles add → 5 Why AC needs both →
6 Could rectangular do everything?

You already have (04b): a phasor is an arrow. **a + jb** says "a across, b up". **M∠θ** says "length M, turned θ from the right". This sheet answers why you keep switching between them.

> **SADIKU, PRACTICE PROBLEM 10.5 (ASSIGNMENT B81)**
> Find current I_o in the circuit of Fig. 10.8 using the superposition theorem.

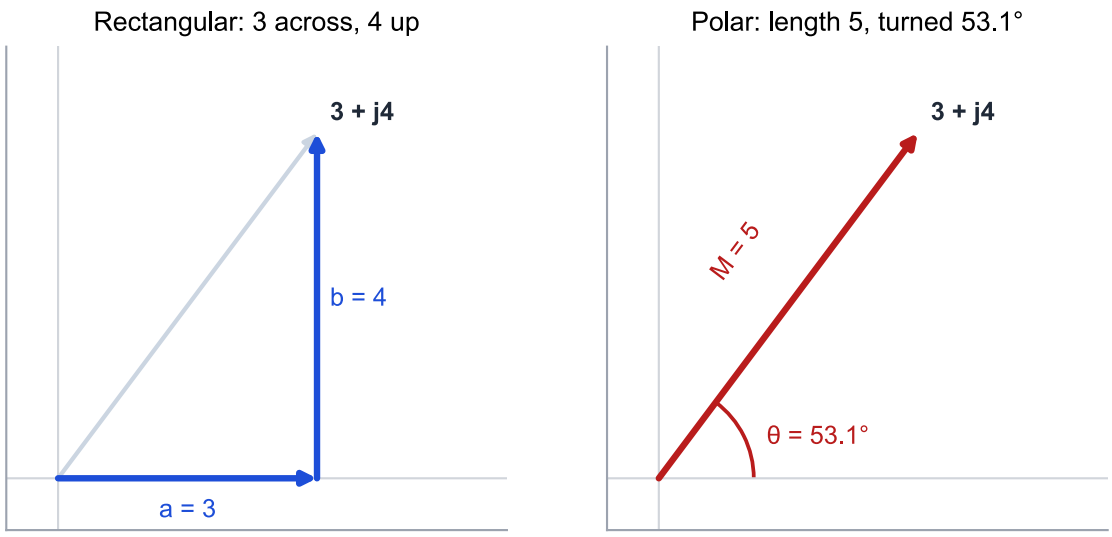
Superposition gave the two pieces: **I_o' = 2.808∠81.84° A** (voltage source alone) and **I_o'' = 3.370∠51.84° A** (current source alone). The last step is $I_o = I_o' + I_o''$.

> **GUESS FIRST**
> Is the answer 6.178∠133.68° (add the sizes, add the angles)? If not, roughly how big is it, and roughly what angle?

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1 · The same arrow, two addresses

Two ways to give directions to the same spot: "walk 3 m east, then 4 m north", or "walk 5 m, facing 53.1° left of east". Same spot. Neither carries more information.



3 + j4 and 5∠53.1° are the same arrow. Conversion (04b): $M = \sqrt{(a^2 + b^2)}$, $\theta = \tan^{-1}(b/a)$; back: $a = M \cos \theta$, $b = M \sin \theta$.

So the question is never "which form is correct". It is **"which form makes this step short"**. Each form has one operation it makes trivial.

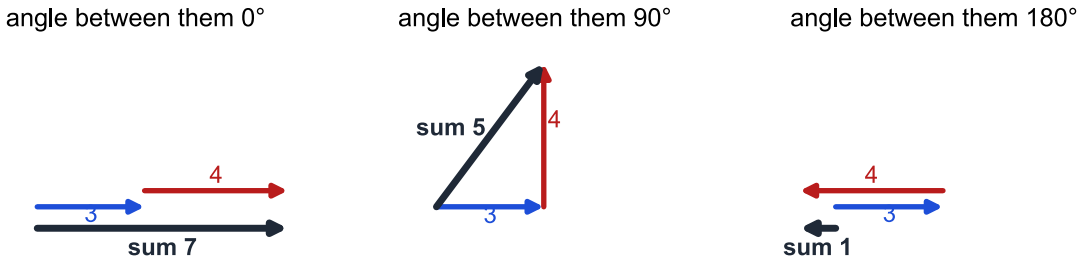
Bridge: the two operations a circuit needs are **adding** (KVL, KCL, series Z) and **multiplying/dividing** ($V = IZ$). Take them one at a time.

2 · Adding: rectangular wins

Adding two arrows means walking one, then the other from where you stopped (head-to-tail). The "across" moves and the "up" moves don't affect each other, so they add separately:

$$(a + jb) + (c + jd) = (a + c) + j(b + d)$$

Polar has no such rule. Two arrows of sizes 3 and 4 can add to anything from 1 to 7. It depends on the angle between them:



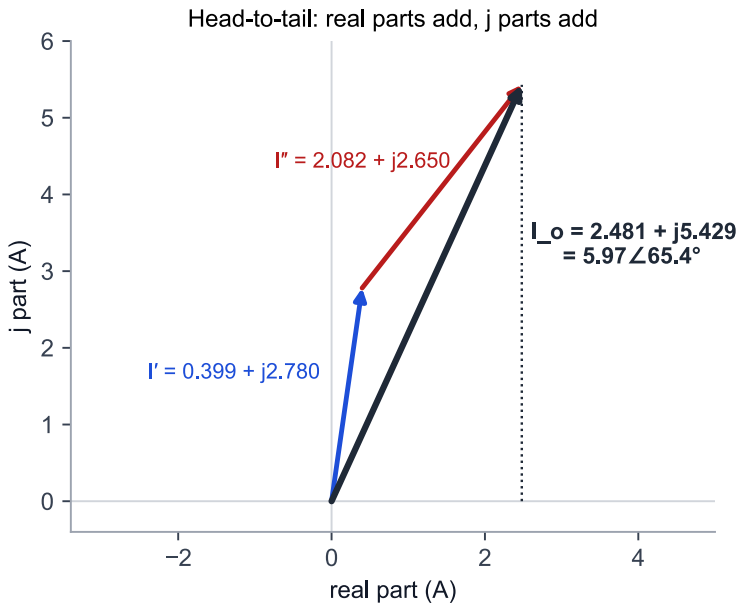
Same sizes, three different answers. In polar you'd need the cosine rule every time.

"Add the sizes" is only true when the angles are equal (left panel). That's the mistake the guess tempted you into.

The anchor, solved. Convert each piece to rectangular, add, convert back:

$$I_o' = 2.808\angle 81.84° = 0.399 + j2.780 \qquad I_o'' = 3.370\angle 51.84° = 2.082 + j2.650$$

$$I_o = 2.481 + j5.429 = 5.97\angle 65.4^\circ \text{ A}$$



The sum's size, 5.97, is less than 2.808 + 3.370 = 6.18, because the two arrows point in different directions. (Book: 5.97∠65.45°, from unrounded pieces.)

CHECK 1 (ANSWERS AT THE END)

(a) Add $10\angle 30^\circ + 10\angle -30^\circ$. Is the result's size 20? (b) Which form did you have to go through?

Bridge: adding is "move". The other thing a circuit does to an arrow is "stretch and turn" (§3).

3 · Multiplying: polar wins

Multiplying by a complex number does two things to an arrow: it **stretches** the length and **turns** the angle.



(5∠53.1°) × (2∠30°): length 5 × 2 = 10, angle 53.1° + 30° = 83.1°.

Polar stores exactly those two things, so the rule is one line each:

multiply: sizes ×, angles + divide: sizes ÷, angles −

Rectangular has to expand brackets and use $j^2 = -1$:

	Rectangular	Polar
$(3 + j4)(1 + j1)$	$3 + j3 + j4 + j^2 4 =$ **$-1 + j7$** (4 products, then regroup)	$5\angle 53.1° \times 1.414\angle 45° =$ **$7.07\angle 98.1°$**
$\frac{10}{3 + j4}$	multiply top and bottom by the conjugate $3 - j4$: $\frac{30 - j40}{25} =$ **$1.2 - j1.6$**	$\frac{10\angle 0°}{5\angle 53.1°} =$ **$2\angle -53.1°$**

Same answers (convert to check). Polar took one step each; rectangular took several, and division needs the **conjugate trick** (flip the sign of the j part to clear j from the bottom).

Multiplying by j is the cleanest case: $j = 1\angle 90°$, so it turns an arrow 90° and leaves its length alone. In rectangular: $j(3 + j4) = -4 + j3$, which is $3 + j4$ turned a quarter-turn.

CHECK 2

Find $\frac{6\angle 0°}{2\angle -90°}$ in polar, then in rectangular.

Bridge: "angles add when you multiply" looks like a convenient rule. It's a consequence of something you already know (§4).

4 · Why angles add: they are exponents

Euler's formula says the polar form is really an exponential:

$$M\angle\theta \qquad \text{is shorthand for} \qquad M\,e^{j\theta} \qquad (e^{j\theta} = \cos\theta + j\sin\theta)$$

Exponents add when you multiply: $10^2 \times 10^3 = 10^5$. In the same way:

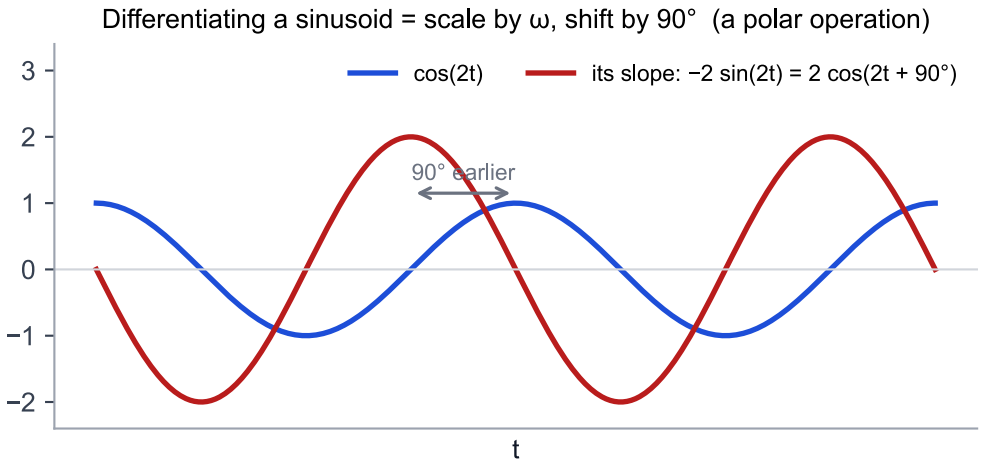
$$M_1 e^{j\alpha} \times M_2 e^{j\beta} = M_1 M_2\, e^{j(\alpha+\beta)}$$

So polar is the **exponent form** of a complex number, and rectangular is the **sum form**. Exponent form makes × and ÷ easy (same reason logarithms turned multiplication into addition on old slide rules). Sum form makes + and − easy. No single form makes both easy. That's a property of the numbers, not a choice of notation.

Bridge: that explains the maths. AC is where you need both operations at once (§5).

5 · Why AC needs both

An inductor's voltage is $L\cdot\frac{di}{dt}$. Differentiate a sinusoid and look at what happens:



The slope of a cosine is the same cosine, scaled by ω and shifted 90° earlier. Scale + turn is multiplication by $\omega\angle 90° = j\omega$.

That's where $Z_L = j\omega L$ comes from. Every R, L and C does a **stretch-and-turn** to a sinusoid. That's **multiplication**, polar's job. Kirchhoff's laws **add** voltages and currents. That's rectangular's job. A circuit has both, so one problem uses both:

Step in an AC problem	Operation	Use
Source $V_m \cos(\omega t + \theta) \rightarrow V_m\angle\theta$	read off size and angle	**polar** (comes that way)
R, $j\omega L$, $-j/(\omega C)$	they're given as a + jb	**rectangular** (comes that way)
Series: $Z_1 + Z_2$	add	**rectangular**
$V = IZ$, $I = V/Z$, dividers	multiply / divide	**polar**
Parallel: $\frac{Z_1 Z_2}{Z_1 + Z_2}$	top ×, bottom +	**both, in one formula**
KVL, KCL, superposition sum	add	**rectangular**
Answer → $M \cos(\omega t + \theta)$	need size and angle	**polar**

In PP 10.5, the 10 A piece needs the divider $I_o'' = 10 \times \frac{Z}{(8 - j2) + Z}$ with $Z = j4 \parallel 6$. Mixed form: $Z = \frac{24\angle 90°}{7.211\angle 33.69°} = 3.328\angle 56.31° = 1.846 + j2.769$; bottom $= 9.846 + j0.769 = 9.876\angle 4.47°$; then $I_o'' = \frac{10 \times 3.328}{9.876}\angle(56.31° - 4.47°) =$ **$3.370\angle 51.84°$**. Each step is in the form where it is one line.

6 · So could rectangular do everything?

Yes. Every step can be done in a + jb and gives the identical answer. Nothing is ever *wrong* in rectangular. It costs:

- **More arithmetic, more slips.** The same I_o'' in rectangular only: two conjugate divisions and about ten multiplications. You get $2.082 + j2.650$ = the same $3.370\angle 51.84°$.
- **A conversion at the end anyway.** The time answer is $M \cos(\omega t + \theta)$, which needs size and angle. Rectangular can't skip polar; it postpones it.
- **Less meaning on the page.** $3.370\angle 51.84°$ tells you "3.37 A peak, 51.84° ahead of the reference". $2.082 + j2.650$ tells you neither until converted.

Polar-only is worse: it has no shortcut for adding, so every KVL or KCL step would need the cosine rule.

THE RULE

Add or subtract → rectangular. Multiply or divide → polar. Convert whenever the next operation changes. "Better" means fewer steps and fewer slips, never "more correct".

QUADRANT TRAP (FROM 04B)

Converting to polar with $\tan^{-1}(b/a)$ is wrong when a is negative: add 180°. $-1 + j7$ is $7.07\angle 98.1°$, not $7.07\angle -81.9°$. Sketch the arrow first.

Summary card

Form	Stores	Easy	Hard
a + jb (sum form)	across, up	+, − (parts add separately)	×, ÷ (brackets, conjugate)
M∠θ = Me$^{j\theta}$ (exponent form)	stretch, turn	×, ÷ (sizes ×÷, angles ±)	+, − (no rule; cosine rule)

Where you use it next: every Sadiku ch. 10–11 problem in the assignment (B80–B127).

Self-test

1. Add (4 + j3) + (4 − j3). Both have size 5: why is the sum's size not 10?
2. Find (2∠30°)(3∠60°) in polar, then in rectangular.
3. Find (4 + j3) ‖ (−j5) using $\frac{Z_1 Z_2}{Z_1 + Z_2}$. Say which form you use for the top, the bottom, and the division.
4. Show in rectangular that multiplying 3 + j4 by j keeps its size 5.
5. For each, which form: (a) series impedances, (b) I = V/Z, (c) the KCL sum at a node, (d) writing i(t)?
6. In one sentence: why does multiplying complex numbers add their angles?

Answers

Guess. No. Sizes add only when angles are equal, and angles never add when you *add* arrows. The real answer is **$5.97\angle 65.4°$ A**: a bit less than 6.18, with an angle between the two pieces' angles (51.84° and 81.84°).

Check 1. (a) $10\angle 30° = 8.660 + j5$, $10\angle -30° = 8.660 - j5$. Sum = **$17.32 + j0 = 17.32\angle 0°$**. Not 20: the arrows are 60° apart. (b) Rectangular.

Check 2. Polar: $\frac{6}{2} = 3$, $0° - (-90°) = 90° \Rightarrow$ **$3\angle 90°$**. Rectangular: **$0 + j3$** (straight up).

Self-test.

1. **$8 + j0 = 8$.** The j parts cancel (one points up, one down). Sizes add only for arrows pointing the same way.
2. $2 \times 3 = 6$, $30° + 60° = 90° \Rightarrow$ **$6\angle 90° = 0 + j6$**.
3. Top (multiply, polar): $(5\angle 36.87°)(5\angle -90°) = 25\angle -53.13°$. Bottom (add, rectangular): $4 + j3 - j5 = 4 - j2 = 4.472\angle -26.57°$. Divide (polar): $\frac{25}{4.472}\angle(-53.13° + 26.57°) =$ **$5.590\angle -26.57° = 5 - j2.5\ \Omega$**.
4. $j(3 + j4) = j3 + j^2 4 =$ **$-4 + j3$**. Size $\sqrt{(16 + 9)} =$ **5**. Same length, turned 90°.
5. (a) **rectangular** (add), (b) **polar** (divide), (c) **rectangular** (add), (d) **polar** (need M and θ).
6. Polar form is $Me^{j\theta}$, and **exponents add when you multiply** powers of the same base.